

```
% direct flights
flight_connects(sanfrancisco, la, 45).

% denver flights
flight_connects(denver, sanfrancisco, 75).
flight_connects(denver, la, 100).

%SD flights
flight_connects(sandiego, la, 45).

% dallas flights
flight_connects(dallas, la, 80).
flight_connects(dallas, sandiego, 90).

% miami flights
flight_connects(miami, dallas, 50).

% NY flights
flight_connects(newyork, chicago, 75).
flight_connects(newyork, denver, 100).
flight_connects(newyork, dallas, 125).
flight_connects(newyork, miami, 90).

% Chicago flights
flight_connects(chicago, denver, 20).
flight_connects(chicago, sanfrancisco, 25).

%route logic part B
% (single route of length 0)
route(C, C, 0).

% direct route
route(A, B, Cost) :-
    flight_connects(A, B, Cost).

% multi route
route(A, B, TotalCost) :-
    flight_connects(A, X, Cost1),
    route(X, B, Cost2),
    TotalCost is Cost1 + Cost2.

% part C
route_of_2_legs(A, B, TotalCost) :-
    flight_connects(A, X, Cost1),
    flight_connects(X, B, Cost2),
    TotalCost is Cost1 + Cost2.
```